CO6602 - Cryptography and Security

Slides designed by: Bismark Asare

University of Chester

Check-in code:

University of Chester

Introduction to Number Theory

Week 2 – Lesson

University of Chester

Objectives

1. Understand the concept of divisibility and the division algorithm.
2. Understand how to use the Euclidean algorithm to find the greatest common divisor.
3. Present an overview of the concepts of modular arithmetic.
4. Discuss key concepts relating to prime numbers.
5. Perform Euler's Totient Function operations.
6. Perform discrete logarithmic operations.

University of Chester

Number Theory

Number theory forms the foundation of cryptographic algorithms. It offers an insight into prime numbers and discrete logarithms.

A number is said to be a prime number if p > 1 and p has no positive divisors except 1 and p.

- Example: $p = 2,3,5,7,11,13$

The numbers which are not prime numbers are referred as composite numbers.

- For any integer n, $n > 1$, let $Z_n = \{0, 1, 2, \ldots, n-1\}$ be a set of numbers. This set is called the set of residues *modulo n*, as the elements are remainders of integers divided by the number n.

- We define the following operations on the set Z_n using the modulo operation.

$$x \oplus_n y = (x + y) \bmod n.$$

$$x \otimes_n y = (xy) \bmod n$$

- Example: $(6 + 7) \bmod 12 = 1$; $5 \times 4 \bmod 12 = 8$;

Clock Arithmetic



University of Chester

Divisibility and the Division Algorithm

Divisibility:

We say that a nonzero b divides a if $a = mb$ for some m, where a, b, and m are integers. That is, b divides a if there is no remainder on division. The notation $b|a$ is commonly used to mean b divides a. Also, if $b|a$, we say that b is a divisor of a.

The remainder r is often referred to as the residue.

University of Chester

Properties of Divisibility of Integers

- If $a|1$, then $a = \pm 1$.
- If $a|b$ and $b|a$, then $a = \pm b$.
- Any $b \neq 0$ divides 0.
- If $a|b$ and $b|c$, then $a|c$:

$$11|66 \text{ and } 66|198 \Longrightarrow 11|198$$

- If $b|g$ and $b|h$, then $b|(mg + nh)$ for arbitrary integers m and n.

To see this last point, note that

- If $b|g$, then g is of the form $g = b \times g_1$ for some integer g_1.
- If $b|h$, then h is of the form $h = b \times h_1$ for some integer h_1.

So

$$mg + nh = mbg_1 + nbh_1 = b \times (mg_1 + nh_1)$$

and therefore b divides $mg + nh$.

$b = 7; g = 14; h = 63; m = 3; n = 2$
$7|14$ and $7|63$.
To show $7|(3 \times 14 + 2 \times 63)$,
we have $(3 \times 14 + 2 \times 63) = 7(3 \times 2 + 2 \times 9)$,
and it is obvious that $7|(7(3 \times 2 + 2 \times 9))$.

University of Chester

The Division Algorithm

Given any positive integer ***n*** and any non-negative integer ***a***, if we divide ***a*** by ***n***, we get an integer quotient ***q*** and an integer remainder ***r*** that obey the following relationship:

$a = qn + r \qquad 0 \leq r < n;\ q = \lfloor a/n \rfloor$

For example:

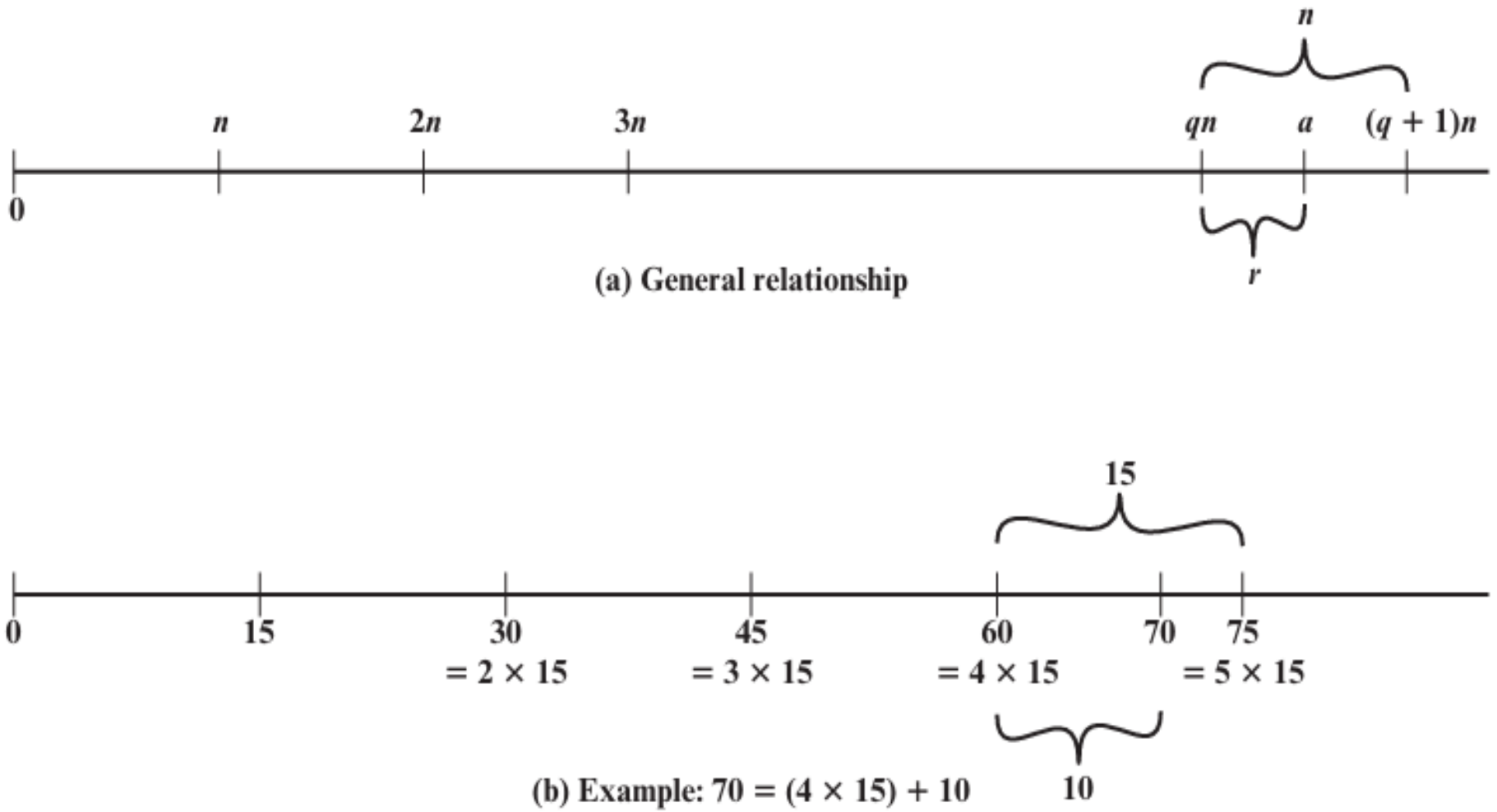
- *a=13, n=3; a/n = 13/3; 4 reminder 'r' 1*
- $q = 4, r = 1$
- $q = \lfloor 13/3 \rfloor$
- $0 \leq r < n;\ \ where\ 'r' = 1, and\ it\ is\ between\ 'r'(3) and\ greater\ than\ 0$

- $a = qn + r; 0 \leq r < n;$
- $13 = 4(3) + 1 \quad ; 0 \leq 1 < 3;\ q = \lfloor 13/3 \rfloor$

$\lfloor x \rfloor$ is the largest integer less than or equal to x

University of Chester

The Division Algorithm



The relationship $a = qn + r;\ 0 \leq r < n;$

University of Chester

Example

$a = qn + r;\ 0 \leq r < n;$

Given:

$a = 11, n = 7;$

$a = qn + r;\ 0 \leq r < n;$

$11 = 1 * 7 + 4;$ $\quad r = 4 \quad q = 1$

$0 \leq r < n: 0 \leq 4 < 7;$

$a = -11, n = 7;$

$a = qn + r;\ 0 \leq r < n;$

$-11 = (-2) * 7 + 3;$ $\quad r = 3 \quad q = -2$

$0 \leq r < n: 0 \leq 3 < 7;$

University of Chester

The Euclidean Algorithm

Euclidean algorithm, a procedure for determining the greatest common divisor (gcd) of two positive integers.

Two integers are ***relatively prime*** if and only if their *only* ***common positive integer factor is 1.***

More formally, the positive integer c is said to be the greatest common divisor of a and b if

1. c is a divisor of a and of b.
2. any divisor of a and b is a divisor of c.

An equivalent definition is the following:

$$\gcd(a, b) = \max[k, \text{such that } k|a \text{ and } k|b]$$

Because we require that the greatest common divisor be positive, $\gcd(a, b) = \gcd(a, -b) = \gcd(-a, b) = \gcd(-a, -b)$. In general, $\gcd(a, b) = \gcd(|a|, |b|)$.

$$\gcd(60, 24) = \gcd(60, -24) = 12$$

University of Chester

Euclidean Algorithm Examples:

Two integers are ***relatively prime*** if and only if their *only* ***common positive integer factor is 1.***

Example 1: 8 and 15

The positive divisors of 8 & 15.

GCD of 8 { 1, 2, 4, 8}

GCD of 15 {1,3,5,15}

The only integer on both list is **1.**

∴ 8 and 15 are relatively prime

Example 2: 10 and 17

The positive divisors of 10 & 17.

GCD of 10 { 1, 2, 5, 10}

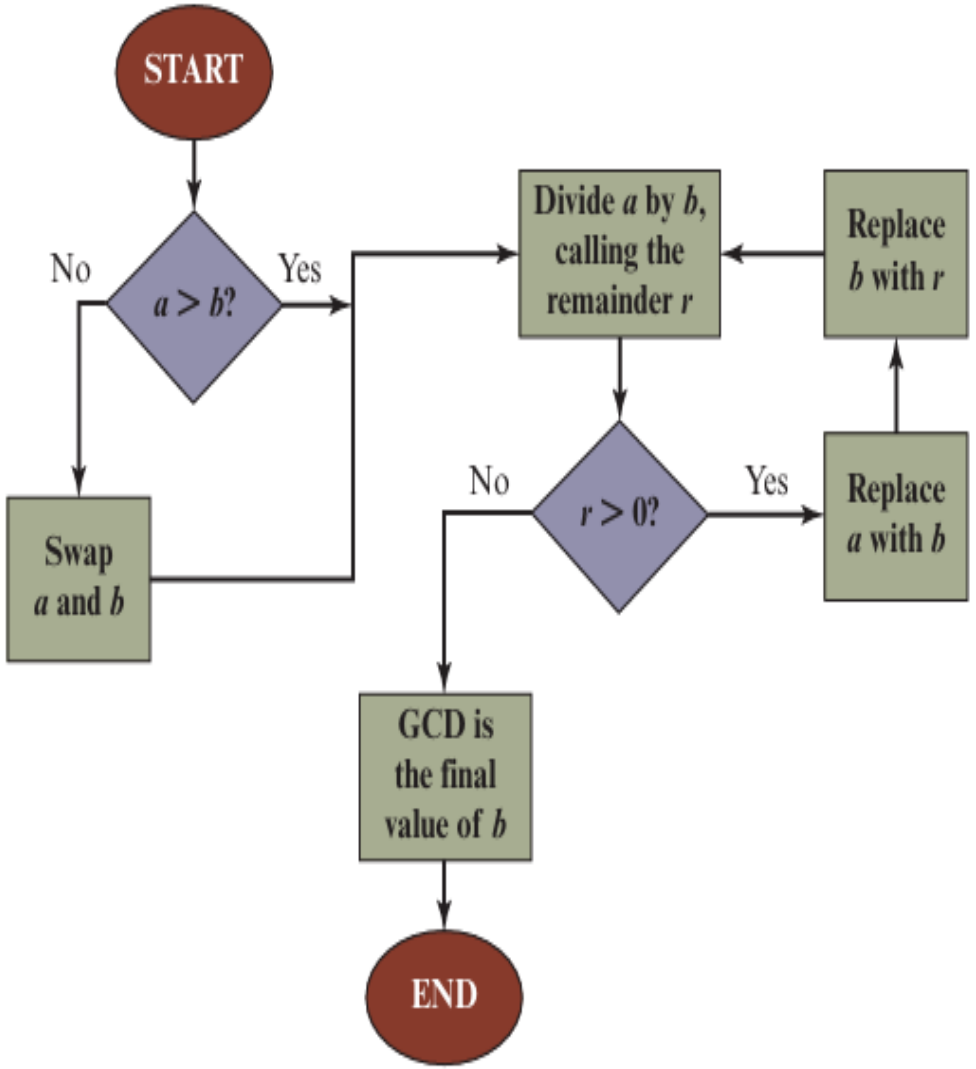
GCD of 17 {1,17}

The only integer on both list is **1.**

∴ 10 and 17 are relatively prime

University of Chester

Euclidean Algorithm



Pseudocode for finding the Greatest Common Divisor **d (GCD)** of **a**, and **b, using the Euclidean Algorithm**

Pseudocode:

Steps:

Compare **a** and **b**, if a < b, then proceed otherwise swap their places.

Divide **a** by **b: (a/b)**

a=q*b + **r**;

where q=quotient, and r=remainder

a/**b** = b***q** + **r**

Step 2: Bring down the previous divisor and remainder.

Divide the previous divisor by the remainder

Step 3: Repeat step 2

Step 4: Repeat step 2 until r = 0, then stop.

Since the remainder has reached 0, the process has completed. The last divisor that resulted in the remainder reaching zero (0) is the **GCD** of **a** and **b**

University of Chester

Euclidean Algorithm

Example: Finding the Greatest Common Divisor (GCD) of 710, and 310

Solution:

Using the division algorithm

a=710; **b**=310:

Step 1: Divide **a** by **b**;710 by 310;

710= (310 *2) + 90

The remainder '**r**' is **90**

Step 2: Bring down the previous divisor and remainder. Divide the previous divisor by the remainder

310 = (3*90) + 40; **r** = 40

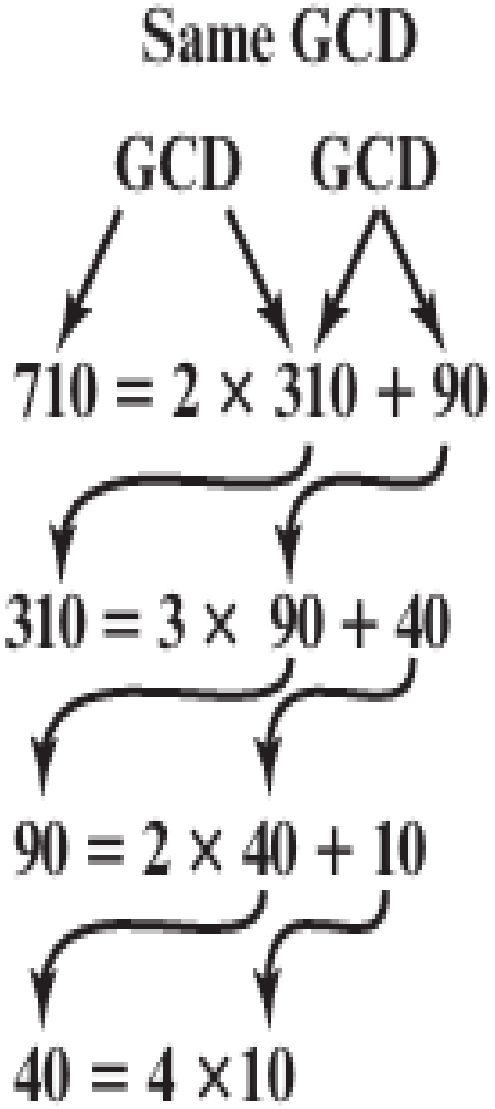
Step 3: Repeat step 2: (90 and 40);

90 = 2*40 + 10; **r** = 10

Step 4: Repeat the previous step until r = 0, then stop. (40 and 10);

40 = (4 * 10) + 0; **r** = 0

Since the remainder has reached 0, the process has completed. The last non-zero remainder is **10**



University of Chester

Example: Euclidean Algorithm

Finding the Greatest Common Divisor (GCD) of 1160718174, and 316258250.

Solution:

- a= 1160718174
- b= 316258250

Steps:

- Compare a and b, if a < b, then proceed otherwise swap their places.
  - a=q*b + **r**;
- where q=quotient, and r=remainder
  - **b/a = q + r**
- (1160718174)/(C) = **q + r**
- (3* 316258250) + 211943424
- Keep dividing the quotient by the lowest value (a) until the remainder 'r' is zero (0), **then the last divisor is the answer.**

To find $d = \gcd(a, b) = \gcd(1160718174, 316258250)$		
$a = q_1 b + r_1$	$1160718174 = 3 \times 316258250 + 211943424$	$d = \gcd(316258250, 211943424)$
$b = q_2 r_1 + r_2$	$316258250 = 1 \times 211943424 + 104314826$	$d = \gcd(211943424, 104314826)$
$r_1 = q_3 r_2 + r_3$	$211943424 = 2 \times 104314826 + 3313772$	$d = \gcd(104314826, 3313772)$
$r_2 = q_4 r_3 + r_4$	$104314826 = 31 \times 3313772 + 1587894$	$d = \gcd(3313772, 1587894)$
$r_3 = q_5 r_4 + r_5$	$3313772 = 2 \times 1587894 + 137984$	$d = \gcd(1587894, 137984)$
$r_4 = q_6 r_5 + r_6$	$1587894 = 11 \times 137984 + 70070$	$d = \gcd(137984, 70070)$
$r_5 = q_7 r_6 + r_7$	$137984 = 1 \times 70070 + 67914$	$d = \gcd(70070, 67914)$
$r_6 = q_8 r_7 + r_8$	$70070 = 1 \times 67914 + 2156$	$d = \gcd(67914, 2156)$
$r_7 = q_9 r_8 + r_9$	$67914 = 31 \times 2156 + 1078$	$d = \gcd(2156, 1078)$
$r_8 = q_{10} r_9 + r_{10}$	$2156 = 2 \times 1078 + 0$	$d = \gcd(1078, 0) = 1078$
Therefore, $d = \gcd(1160718174, 316258250) = 1078$		

University of Chester

Euclidean Algorithm Example

Dividend	Divisor	Quotient	Remainder
$a = 1160718174$	$b = 316258250$	$q_1 = 3$	$r_1 = 211943424$
$b = 316258250$	$r_1 = 211943434$	$q_2 = 1$	$r_2 = 104314826$
$r_1 = 211943424$	$r_2 = 104314826$	$q_3 = 2$	$r_3 = 3313772$
$r_2 = 104314826$	$r_3 = 3313772$	$q_4 = 31$	$r_4 = 1587894$
$r_3 = 3313772$	$r_4 = 1587894$	$q_5 = 2$	$r_5 = 137984$
$r_4 = 1587894$	$r_5 = 137984$	$q_6 = 11$	$r_6 = 70070$
$r_5 = 137984$	$r_6 = 70070$	$q_7 = 1$	$r_7 = 67914$
$r_6 = 70070$	$r_7 = 67914$	$q_8 = 1$	$r_8 = 2156$
$r_7 = 67914$	$r_8 = 2156$	$q_9 = 31$	$r_9 = 1078$
$r_8 = 2156$	$r_9 = 1078$	$q_{10} = 2$	$r_{10} = 0$

University of Chester

Modular Arithmetic

The Modulus:

If ***a*** is an integer and ***n*** is a positive integer, we define ***a*** mod ***n*** to be the remainder when ***a*** is divided by ***n***. The integer ***n*** is called the modulus.

$$a = qn + r \qquad 0 \leq r < n; q = \lfloor a/n \rfloor$$

$$a = \lfloor a/n \rfloor \times n + (a \bmod n)$$

Where:

a=Divident; b=Divisor; q=Quotient; n=Modulus; r=Remainder

Example:

21 mod 5; where a = 21, n = 5

- a = (q*n) + r
- 21 = (4*5) + 1

21 mod 5 is **1**

University of Chester

Modulo Example:

Find -11 mod 7

Solution:

The answer for a modulus, should be a number from 0 to one less than that number.

For example: Mod 7 = (0, 1, 2, 3, 4, 5, 6)

- To find **a negative number** modulus, keep adding the negative number to the modulus until you get a positive number then you stop.

$$a = qn + r \qquad 0 \le r < n; q = \lfloor a/n \rfloor$$

$$a = \lfloor a/n \rfloor \times n + (a \bmod n)$$

$$11 \bmod 7 = 4; \qquad -11 \bmod 7 = 3$$

Step 1:

- - 11 + 7 = -4

Step 2:

- - 4 + 7 = 3

∴ - 11 mod 7 is **3**

University of Chester

Further Examples:

Can you solve the following:

1. 81 (mod 17)
2. 120 (mod 19)
3. -13 (mod 6)
4. -17 (mod 5)
5. -15 (mod 2)
6. -4 (mod 8)

University of Chester

Congruent Modulus (≡)

Two integers ***a*** and ***b*** are said to be congruent modulo ***n***, if (***a*** mod ***n***) = (***b*** mod n). This is written as ***a*** ≡ ***b*** $(mod\ n)^2$

Note that if $a \equiv 0\ (mod\ n), then\ n|a$

If ***a*** congruent to **0** mod ***n***, then ***n*** is divisible by ***a.***

Example 1:

73 and 4 are congruent mod 23.

73 mod 23 = (3 * 23) + 4

73 mod 23 = 4

4 mod 23 = 4

∴ 73 ≡ 4 mod 23 // 73 and 4 will leave the same reminder when using modulus 23.

21 and - 9 are congruent mod 10

21 mod 10 = (2*10) + 1

21 mod 10 = 1

-9 mod 10 = (10 + (-9)) = 1

∴ 21 ≡ -9 mod 10 // 21 and -9 will leave the same reminder when using modulus 10

University of Chester

Properties of Congruences

Congruences have the following properties:

1. $a \equiv b \,(mod\ n), if\ n \mid (a - b)$.
2. $a \equiv b \,(mod\ n), implies\ b \equiv a \,(mod\ n)$.
3. $a \equiv b \,(mod\ n)\ and\ b \equiv c \,(mod\ n)\ imply\ a \equiv c \,(mod\ n)$

Examples:

23 and 8 are congruent (mod 5)

23 (mod 5) = (4 * 5) + 3 ; 23 (mod 5) = 3

8 (mod 5) = (1 * 5) + 3 ; 8 (mod 5) = 3

University of Chester

Further Examples

Can you solve by proofing that these congruent expressions are true?

i. 5 and -11 are congruent (mod 8)

$5 \equiv -11 \pmod 8$

ii. 81 and 0 are congruent (mod 27)

$81 \equiv 0 \pmod{27}$

iii. 50 and -13 are congruent (mod 9)

$50 \equiv -13 \pmod 9$

University of Chester

Modular Arithmetic Operations

Modular arithmetic exhibits the following properties:

1. $[(a \bmod n) + (b \bmod n)] \bmod n = (a + b) \bmod n$
2. $[(a \bmod n) - (b \bmod n)] \bmod n = (a - b) \bmod n$
3. $[(a \bmod n) \times (b \bmod n)] \bmod n = (a \times b) \bmod n$

$11 \bmod 8 = 3; 15 \bmod 8 = 7$

$[(11 \bmod 8) + (15 \bmod 8)] \bmod 8 = 10 \bmod 8 = 2$

$(11 + 15) \bmod 8 = 26 \bmod 8 = 2$

$[(11 \bmod 8) - (15 \bmod 8)] \bmod 8 = -4 \bmod 8 = 4$

$(11 - 15) \bmod 8 = -4 \bmod 8 = 4$

$[(11 \bmod 8) \times (15 \bmod 8)] \bmod 8 = 21 \bmod 8 = 5$

$(11 \times 15) \bmod 8 = 165 \bmod 8 = 5$

University of Chester

Properties of Modular Arithmetic - Addtion

+	0	1	2	3	4	5	6	7
0	0	1	2	3	4	5	6	7
1	1	2	3	4	5	6	7	0
2	2	3	4	5	6	7	0	1
3	3	4	5	6	7	0	1	2
4	4	5	6	7	0	1	2	3
5	5	6	7	0	1	2	3	4
6	6	7	0	1	2	3	4	5
7	7	0	1	2	3	4	5	6

(a) Addition modulo 8

Define the set Z_n as the set of nonnegative integers less than n:

$$Z_n = \{0, 1, \ldots, (n-1)\}$$

University of Chester

Properties of Modular Arithmetic - Multiplication

×	0	1	2	3	4	5	6	7
0	0	0	0	0	0	0	0	0
1	0	1	2	3	4	5	6	7
2	0	2	4	6	0	2	4	6
3	0	3	6	1	4	7	2	5
4	0	4	0	4	0	4	0	4
5	0	5	2	7	4	1	6	3
6	0	6	4	2	0	6	4	2
7	0	7	6	5	4	3	2	1

(b) Multiplication modulo 8

Define the set Z_n as the set of nonnegative integers less than n:

$$Z_n = \{0, 1, \ldots, (n-1)\}$$

University of Chester

Properties of Modular Arithmetic for Integers in Z_n

Property	Expression
Commutative Laws	$(w + x) \bmod n = (x + w) \bmod n$ $(w \times x) \bmod n = (x \times w) \bmod n$
Associative Laws	$[(w + x) + y] \bmod n = [w + (x + y)] \bmod n$ $[(w \times x) \times y] \bmod n = [w \times (x \times y)] \bmod n$
Distributive Law	$[w \times (x + y)] \bmod n = [(w \times x) + (w \times y)] \bmod n$
Identities	$(0 + w) \bmod n = w \bmod n$ $(1 \times w) \bmod n = w \bmod n$
Additive Inverse $(-w)$	For each $w \in Z_n$, there exists a z such that $w + z \equiv 0 \bmod n$

University of Chester

Prime Numbers

An integer ***p***>1 is a prime number if and only if its only divisors are ±1 and ±***p.***

All numbers other than ±1 and the prime numbers are **composite numbers.**

Examples of Prime Numbers:

{2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97, ….}

University of Chester

Prime factors

$12 = 2^2 * 3$

$18 = 2 * 3^2$

$36 = 2^2 * 3^2$

$100 = 2^2 * 5^2$

$300 = 2^2 * 3 * 5^2$

University of Chester

Fermat's and Euler Theorems

These two theorems play a crucial roles in public-key cryptography.

Fermat's Theorem:

If ***p*** is prime and ***a*** is a positive integer not divisible by ***p***, then:

$$a^{p-1} \equiv 1 \ (mod \ p)$$

An alternative form of Fermat's theorem is also useful: If ***p*** is prime and ***a*** is a positive integer, then

$$a^{p} \equiv a \ (mod \ p)$$

University of Chester

Fermat's Theorem

$$a^{p-1} \equiv 1 \ (mod \ p)$$

a = 7, p = 19

$a^{p-1} \ (mod \ p)$

$7^{19-1} \ (mod \ 19)$

$a^{p-1} \ (mod \ p) = 7^{18} \ (mod \ 19)$ = $7^{16} * 7^{2} \ (mod \ 19)$

$7^{2} \ (mod \ 19)$ = $49 \ (mod \ 19)$; (2*19) + 11; = 11 (mod 19)

$7^{4} \ (mod \ 19)$ = $11 * 11 \ (mod \ 19)$; 121 (mod 19) = 7 (mod 19)

$7^{8} \ (mod \ 19)$ = $7 * 7 \ (mod \ 19)$; 49 (mod 19) = 11 (mod 19)

$7^{16} \ (mod \ 19)$ = $11 * 11 \ (mod \ 19)$; 121 (mod 19) = 7 (mod 19)

$a^{p-1} \ (mod \ p) = 7^{18}$ = $7^{16} * 7^{2} \equiv 7 * 11 \equiv \mathbf{1} \ (mod \ 19)$

University of Chester

Example of Fermat's Theorem

$a = 3$, $p = 5$

$\boldsymbol{a^{p-1}\ (mod\ p)}$

$a^{5-1}\ (mod\ 5)$

$a^{p-1}\ (mod\ p) = 3^{5-1}\ (mod\ 5)$ = $3^4\ (mod\ 5)$

$3^4\ (mod\ 5)$ = $3^2 * 3^2\ (mod\ 5)$

$3^2\ (mod\ 5)$ = $9\ (mod\ 5)$; 1*5 + 4 (mod 5) = 4 (mod 5)

$3^4\ (mod\ 5)$ = $4 * 4\ (mod\ 5)$; 16 (mod 5)

$3^4\ (mod\ 5)$ = $16\ (mod\ 19)$; 3*5 + 1 (mod 5) = 1 (mod 5)

$a^{p-1}\ (mod\ p) = 3^4 = 3^2 * 3^2 \equiv 4 * 4 \equiv \mathbf{1}\ (mod\ 5)$

University of Chester

Example of Fermat's Theorem

Alternative form of Fermat's Theorem:

$p = 5, a = 3$

$$\boldsymbol{a^p \equiv a (mod\ p)}$$

$$3^5 \equiv 3 (mod\ 5)$$

$$3^5 \equiv 3^2 * 3^3 \equiv 3 (mod\ 5)$$

$3^2\ (mod\ 5) = 3^2\ (mod\ 5)$; 4 (mod 5)

$3^3\ (mod\ 5) = 3^2 * 3\ (mod\ 5)$; 4*3 (mod 5) = 12 (mod 5) ≡ 2 (mod 5)

$$3^5 \equiv 3^2 * 3^3 = 2 * 4\ (mod\ 5) \equiv 3$$

$$\boldsymbol{a^p \equiv a (mod\ p)}; 3^5 = 3^2 * 3^3 \equiv 3 (mod\ 5)$$

University of Chester

Euler's Totient Function

Euler's totient function: φ(n), is defined as the number of positive integers less than ***n*** and relatively prime to ***n***. By convention, φ(1) = 1.

Determine φ(37) and φ(35)

Because 37 is prime, all of the positive integers from 1 through 36 are relatively prime to 37, Thus, φ(37) = 36

To determine φ(35), we list all of the positive integers less than 35 that are relatively prime to it.

{1,2,3,4,6,8,9,11,12,13,16,17,18,19,22,23,24,26,27,29,31,32,33,34}

There are 24 numbers on the list, so φ(35) = 24

University of Chester

Examples of Euler's Totient

Φ(p) = p – 1

4 = {1, 3}

5 = {1, 2, 3, 4}

6 = {1, 5}

7 = {1, 2, 3, 4, 5, 6}

8 = {1, 3, 5, 7}

9 = {1, 2, 4, 5, 7, 8}

10 = {1, 3, 7, 9}

n	$\phi(n)$
1	1
2	1
3	2
4	2
5	4
6	2
7	6
8	4
9	6
10	4

n	$\phi(n)$
11	10
12	4
13	12
14	6
15	8
16	8
17	16
18	6
19	18
20	8

n	$\phi(n)$
21	12
22	10
23	22
24	8
25	20
26	12
27	18
28	12
29	28
30	8

$$\phi(n) = \phi(pq) = \phi(p) \times \phi(q) = (p - 1) \times (q - 1)$$

University of Chester

The Chinese Remainder Theorem

The **Chinese Remainder Theorem (CRT)** is a powerful mathematical tool used to solve a **system of simultaneous linear congruences** (a set of remainder puzzles).

It offers fundamental application for modern digital communication.

Application of CRT:

1. Cryptography (RSA)
2. Computer Arithmetic
3. Secret Sharing

University of Chester

Discrete Logarithms

- Discrete logarithms are fundamental to a number of public-key algorithms, including Diffie–Hellman key exchange and the digital signature algorithm (DSA).

- The power of an integer, Modulo n

$$a^{\Phi(n)} \equiv 1 \ (mod\ n)$$
$$a^{m} \equiv 1 \ (mod\ n)$$

University of Chester

The Powers of Integers, Modulo 19

a	a^2	a^3	a^4	a^5	a^6	a^7	a^8	a^9	a^{10}	a^{11}	a^{12}	a^{13}	a^{14}	a^{15}	a^{16}	a^{17}	a^{18}
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
2	4	8	16	13	7	14	9	18	17	15	11	3	6	12	5	10	1
3	9	8	5	15	7	2	6	18	16	10	11	14	4	12	17	13	1
4	16	7	9	17	11	6	5	1	4	16	7	9	17	11	6	5	1
5	6	11	17	9	7	16	4	1	5	6	11	17	9	7	16	4	1
6	17	7	4	5	11	9	16	1	6	17	7	4	5	11	9	16	1
7	11	1	7	11	1	7	11	1	7	11	1	7	11	1	7	11	1
8	7	18	11	12	1	8	7	18	11	12	1	8	7	18	11	12	1
9	5	7	6	16	11	4	17	1	9	5	7	6	16	11	4	17	1
10	5	12	6	3	11	15	17	18	9	14	7	13	16	8	4	2	1
11	7	1	11	7	1	11	7	1	11	7	1	11	7	1	11	7	1
12	11	18	7	8	1	12	11	18	7	8	1	12	11	18	7	8	1
13	17	12	4	14	11	10	16	18	6	2	7	15	5	8	9	3	1
14	6	8	17	10	7	3	4	18	5	13	11	2	9	12	16	15	1
15	16	12	9	2	11	13	5	18	4	3	7	10	17	8	6	14	1
16	9	11	5	4	7	17	6	1	16	9	11	5	4	7	17	6	1
17	4	11	16	6	7	5	9	1	17	4	11	16	6	7	5	9	1
18	1	18	1	18	1	18	1	18	1	18	1	18	1	18	1	18	1

University of Chester

Properties of Logarithms

1. $y = x^{\log_x(y)}$
2. $\log_x(1) = 0$
3. $\log_x(x) = 1$
4. $\log_x(yz) = \log_x(y) + \log_x(z)$
5. $\log_x(y^r) = r * \log_x(y)$

University of Chester

Calculation of Discrete Logarithm

$y = g^x \bmod p$

Given g, x, and p, it is a straightforward matter to calculate y.

University of Chester

Questions

University of Chester